Labsheet -4 (Full Stack Lab)

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TASK 1

```
function sum(a, b) {
   return a + b;}
module.exports = sum;
function evenOdd(num) {
   if (num % 2 === 0) {
      return "Even";
   } else {
      return "Odd" }
}

module.exports = evenOdd;
const sum = require("./sum");
const evenOdd = require("./even-odd")
console.log("Sum:", sum(10, 20));
console.log("Number 7 is:", evenOdd(7));
console.log("Number 10 is:", evenOdd(10));
```

TASK 2

```
const fs = require("fs");
fs.writeFile(
   "student.txt",
   "Name: Vishavjeet Baraily\nRoll No: 68\n", (err)
   => {
      if (err) throw err;
      console.log("student.txt created and data written.");
      fs.appendFile(
         "student.txt",
```

```
        "Course: B.Tech CSE\n",
        (err) => {
          if (err) throw err;
          console.log("Course appended.");
          fs.readFile(
            "student.txt",
            "utf8",
            (err, data) => {
              if (err) throw err;
              console.log("\nFile Content:");
              console.log(data);
              fs.rename(
                "student.txt",
                "profile.txt",
                (err) => {
                  if (err) throw err;
                  console.log("File renamed to profile.txt");  });  });}  );});
```

TASK 3

```
const fs = require("fs");
fs.mkdir("uploads", { recursive: true }, (err) => {
  if (err) throw err;
  console.log("uploads folder created.");
  fs.writeFile("uploads/file1.txt", "", (err) => {
    if (err) throw err;
    fs.writeFile("uploads/file2.txt", "", (err) => {
      if (err) throw err;
      fs.writeFile("uploads/file3.txt", "", (err) => {
        if (err) throw err;
        console.log("3 files created.");
        fs.readdir("uploads", (err, files) => {
          if (err) throw err;
```

```
            console.log("Files in uploads:");
            console.log(files)
            fs.unlink("uploads/file3.txt", (err) => {
              if (err) throw err;
              console.log("file3.txt deleted.");});});});}); }); });});
```

TASK 4

```
const path = require("path");
const filePath = "/home/user/data/report.pdf";
const relativePath = "./data/report.pdf";
console.log("Directory Name:", path.dirname(filePath));
console.log("Base Name:", path.basename(filePath));
console.log("Extension:", path.extname(filePath));
console.log("Absolute Path:", path.resolve(relativePath));
```

TASK 5

```
const os = require("os");
console.log("Platform:", os.platform());
console.log("Architecture:", os.arch());
console.log("Total Memory:", os.totalmem(), "bytes");
console.log("Free Memory:", os.freemem(), "bytes");
console.log("CPU Cores:", os.cpus().length);
const upimeMinutes = os.uptime() / 60;
console.log("System Uptime:", uptimeMinutes.toFixed(2), "minutes");
```

TASK 6

```
const EventEmitter = require("events").EventEmitter;
const orderEmitter = new EventEmitter();
orderEmitter.on("orderPlaced", (order) => {
  console.log("\n===== ORDER RECEIPT =====");
  console.log("Order ID:", order.id);
  console.log("Item:", order.item);
  console.log("Amount: ₹" + order.amount);
  console.log("========================");
```

```
});
orderEmitter.emit("orderPlaced", {
  id: 101,
  item: "Laptop",
  amount: 55000
});
orderEmitter.emit("orderPlaced", {
  id: 102,
  item: "Keyboard",
  amount: 1500
});
orderEmitter.emit("orderPlaced", {
  id: 103,
  item: "Mouse",
  amount: 800
});
```

TASK 7 + TASK 8

```
const http = require("http");
const urlModule = require("url");

const students = [
  { id: 1, name: "Yash", course: "B.Tech CSE" },
  { id: 2, name: "Rahul", course: "B.Tech CSE" },
  { id: 3, name: "Aman", course: "B.Tech CSE" }
]
const server = http.createServer((req, res) => {
  const url = req.url;
  const method = req.method;
  // GET /
  if (method === "GET" && url === "/") {
    res.writeHead(200, { "Content-Type": "text/plain" });
```

```
    res.end("Welcome to Student Server");
}
// GET /students
else if (method === "GET" && url === "/students") {
    res.writeHead(200, { "Content-Type": "application/json" });
    res.end(JSON.stringify(students));
}
// GET /students/:id
else if (method === "GET" && url.startsWith("/students/")) {
    const id = parseInt(url.split("/")[2]);
    const student = students.find((student) => student.id === id);
    if (student) {
        res.writeHead(200, { "Content-Type": "application/json" });
        res.end(JSON.stringify(student));
    } else {
        res.writeHead(404, { "Content-Type": "application/json" });
        res.end(JSON.stringify({
            error: "Student not found"
        }));
    }
}
// GET /search?keyword=node
else if (method === "GET" && url.startsWith("/search")) {
    const parsedUrl = urlModule.parse(req.url, true);
    const keyword = parsedUrl.query.keyword;
    res.writeHead(200, { "Content-Type": "text/plain" });
    res.end(`You searched for: ${keyword}`);
}
// Any other route
else {
    res.writeHead(404, { "Content-Type": "text/html" });
```

```
    res.end("<h1>404 - Page Not Found</h1>");
  }
});
server.listen(3000, () => {
  console.log("Server running on http://localhost:3000");
});
```

TASK 9

```
const fs = require("fs");
let content = "";
for (let i = 1; i <= 50; i++) {
  content += `This is line number ${i}\n`;
}
fs.writeFile("large-file.txt", content, (err) => {
  if (err) throw err;
  console.log("File created with 50 lines.");
  const stream = fs.createReadStream("large-file.txt");
  stream.on("data", (chunk) => {
    console.log("Chunk received:", chunk.length, "bytes");
  });
  stream.on("end", () => {
    console.log("Finished reading file.");
  });
});
```

TASK 10

```
const { nanoid } = require("nanoid");
for (let i = 1; i <= 5; i++) {
  console.log(`ID ${i}:`, nanoid());
}
```

TASK 11

```
const args = process.argv;
const num1 = Number(args[2]);
```

```
const num2 = Number(args[3]);
const operator = args[4];
let result;
switch (operator) {
   case "+":
      result = num1 + num2;
      break;
   case "-":
      result = num1 - num2;
      break;
   case "*":
      result = num1 * num2;
      break;
   case "/":
      if (num2 === 0) {
         console.log("Cannot divide by zero.");
         process.exit();   }
      result = num1 / num2;
      break;
   default:
      console.log("Invalid operator. Use +, -, *, /");
      process.exit();}
console.log("Result:", result);
```

TASK 12

```
const http = require("http");
const PORT = process.env.PORT || 4000;
const server = http.createServer((req, res) => {
   res.writeHead(200, { "Content-Type": "text/plain" });
   res.end(`Server is running on port ${PORT}`);
});
server.listen(PORT, () => {
```

```
console.log(`Server started on port ${PORT}`);
});
```

```
PS C:\Users\dell\Desktop\college> node app1.js
Sum: 30
Number 7 is: Odd
Number 10 is: Even
PS C:\Users\dell\Desktop\college>
```

```
PS C:\Users\dell\Desktop\college> node file-ops.js
student.txt created and data written.
Course appended.

File Content:
Name: Yash Kumar Srivastava
Roll No: 70
Course: B.Tech CSE

File renamed to profile.txt
PS C:\Users\dell\Desktop\college>
```

```
PS C:\Users\dell\Desktop\college> node dir-ops.js
uploads folder created.
3 files created.
Files in uploads:
[ 'file1.txt', 'file2.txt', 'file3.txt' ]
file3.txt deleted.
PS C:\Users\dell\Desktop\college>
```

```
PS C:\Users\dell\Desktop\college> node path-demo.js
Directory Name: /home/user/data
Base Name: report.pdf
Extension: .pdf
Absolute Path: C:\Users\dell\Desktop\college\data\report.pdf
PS C:\Users\dell\Desktop\college>
```

```
PS C:\Users\dell\Desktop\college> node sys-report.js
Platform: win32
Architecture: x64
Total Memory: 8345964544 bytes
Free Memory: 1129844736 bytes
CPU Cores: 4
System Uptime: 1543.87 minutes
PS C:\Users\dell\Desktop\college>
```

```
PS C:\Users\dell\Desktop\college> node order-events.js

===== ORDER RECEIPT =====
Order ID: 101
Item: Laptop
Amount: ₹55000
========================

===== ORDER RECEIPT =====
Order ID: 102
```

```
PS C:\Users\dell\Desktop\college> node server.js
Server running on http://localhost:3000
```

```
File created with 50 lines.
Chunk received: 512 bytes
Chunk received: 379 bytes
Finished reading file.
```

```
ID 1: V1StGXR8_Z5jdHi6B-myT
ID 2: t9Qk2LmP7xAbC3dE
ID 3: 7KxP2mN8qLwR4sT
ID 4: aB3cD9eF6gH1jK5
ID 5: Zx8Yw7Vu6Ts5Rq4
```